Developing and Applying Explainable AI Using Computational Models of Argument

Professor Katie Atkinson,
School of Computer Science and Informatics.



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“No greater misfortune could happen to anyone than that of developing a dislike for argument”

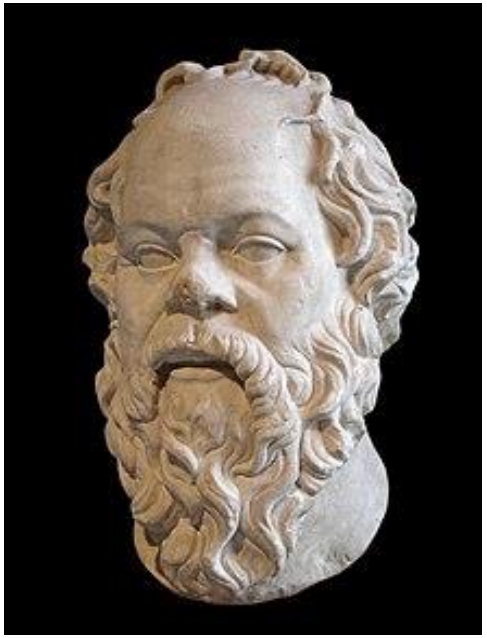


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Outline



- Computational models of argument
- Modelling argumentation for legal reasoning
- My favourite legal case
- Building AI tools for deciding legal case outcomes
- Explainable AI for chemistry applications
- Concluding thoughts

What is an argument?

Argumentation scheme for expert opinion:

Major premise: Source *E* is an expert in the subject domain *D* containing proposition *A*.

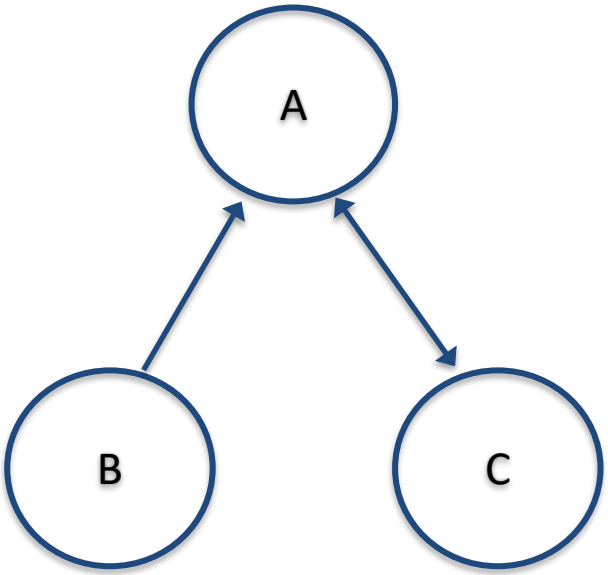
Minor premise: *E* asserts that proposition *A* is true.

Conclusion: *A* may be plausibly be taken to be true.

Associated critical questions:

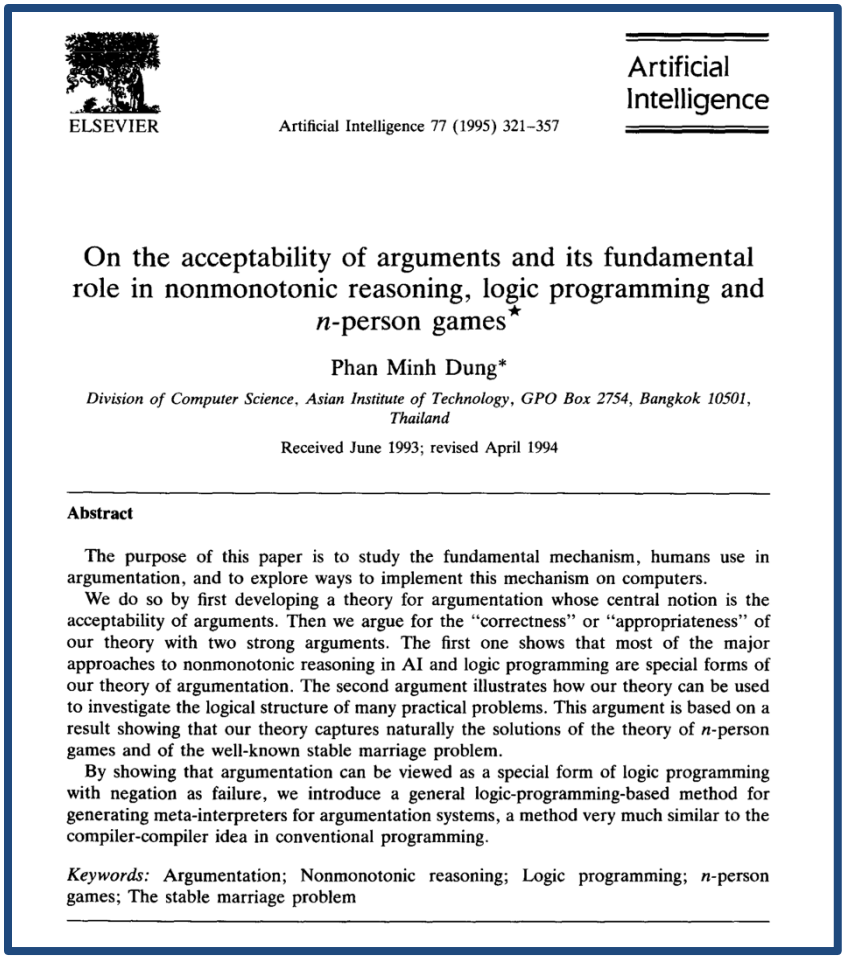
- How credible is *E* as an expert source?
- Is *E* an expert in the field that *A* is in?
- What did *E* assert that implies *A*?
- Is *E* personally reliable as a source?
- Is *A* consistent with what other experts assert?
- Is *E*'s assertion based on evidence?

What is an argument?



Computational Models of Argument

- Argumentation: the exchange of arguments concerning matters of belief or matters of action
- Computational models of argument are used to model debates involving competing arguments to enable the debates to be executed in a computer program
- A seminal paper in the literature on computational models of argument:



ELSEVIER

Artificial Intelligence 77 (1995) 321–357

Artificial Intelligence

On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games*

Phan Minh Dung*

Division of Computer Science, Asian Institute of Technology, GPO Box 2754, Bangkok 10501, Thailand

Received June 1993; revised April 1994

Abstract

The purpose of this paper is to study the fundamental mechanism, humans use in argumentation, and to explore ways to implement this mechanism on computers.

We do so by first developing a theory for argumentation whose central notion is the acceptability of arguments. Then we argue for the "correctness" or "appropriateness" of our theory with two strong arguments. The first one shows that most of the major approaches to nonmonotonic reasoning in AI and logic programming are special forms of our theory of argumentation. The second argument illustrates how our theory can be used to investigate the logical structure of many practical problems. This argument is based on a result showing that our theory captures naturally the solutions of the theory of n-person games and of the well-known stable marriage problem.

By showing that argumentation can be viewed as a special form of logic programming with negation as failure, we introduce a general logic-programming-based method for generating meta-interpreters for argumentation systems, a method very much similar to the compiler-compiler idea in conventional programming.

Keywords: Argumentation; Nonmonotonic reasoning; Logic programming; n-person games; The stable marriage problem

Abstract Argumentation Frameworks

- *An argumentation framework (AF) is a pair H = <X, A> in which X is a finite set of arguments and A ⊂ X × X is the attack relationship for H.*

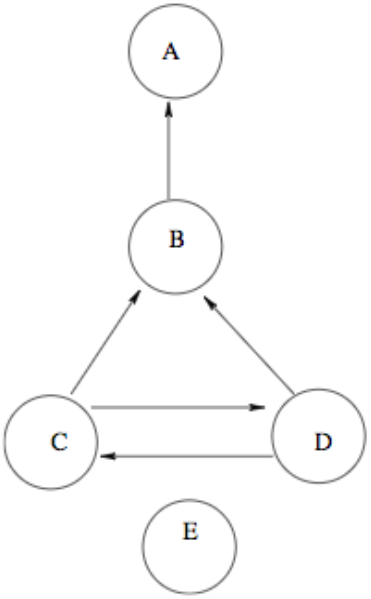
  <x,y> ∈ A denotes that 'y is attacked by x' or 'x attacks y'.

- *x ∈ X is acceptable with respect to a set S if for every y ∈ X that attacks x there is some z ∈ S that attacks y.*

- *S is conflict-free if no argument in S is attacked by any other argument in S.*

- *A set of arguments S, is admissible if it is conflict-free and each of its members is acceptable w.r.t. S.*

- Admissible sets provide the basic context for examining ideas of "maximally consistent sets of beliefs".



Computational Models of Argument

- Computational models of argument can be applied in many domains
  - My focus: law
  - More recently: chemistry



- Arguments are exchanged in a wide variety of scenarios when undertaking legal work
  - A key task is deciding the outcome of legal cases: who should win the case, and why?
  - Walkthrough example…

Popov v. Hayashi

- Baseball player Barry Bonds hit a record-breaking 73rd home run in the 2001 season
- Mr Popov caught it in the upper part of the webbing of his baseball glove – a snowcone catch
- But a mob threw Popov to the ground, the ball dislodged from his glove and was picked up by Mr Hayashi

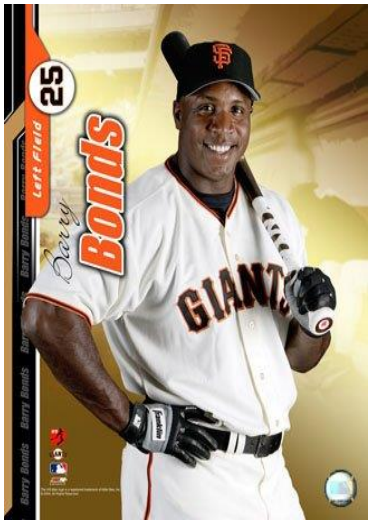


Image: sports-central.org



Image: Amazon: Wonninek



Image: Pinterest: Buzzfeed



Image: Marcio Jose Sanchez / AP

Who owns the baseball?

- The Honourable Kevin M. McCarthy decided in the California Superior Court

- A film was also made about the case: *Up for Grabs*
  - See trailer on YouTube



Image: Facebook:SCSCOURT

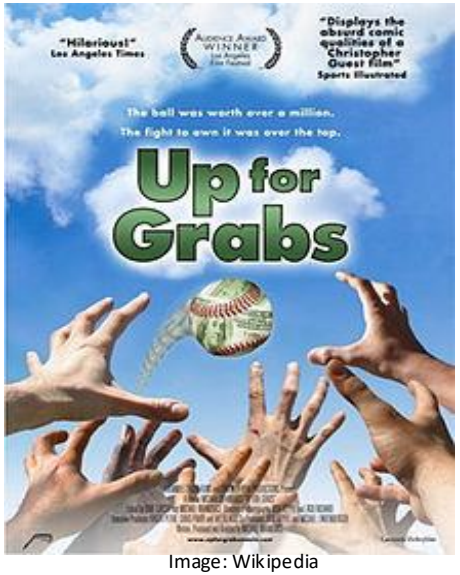


Image: Wikipedia



Up for Grabs movie. Trailer on YouTube at: https://www.youtube.com/watch?v=NP8j_X88bSI

Can an AI program come to the same decision that the judge did?

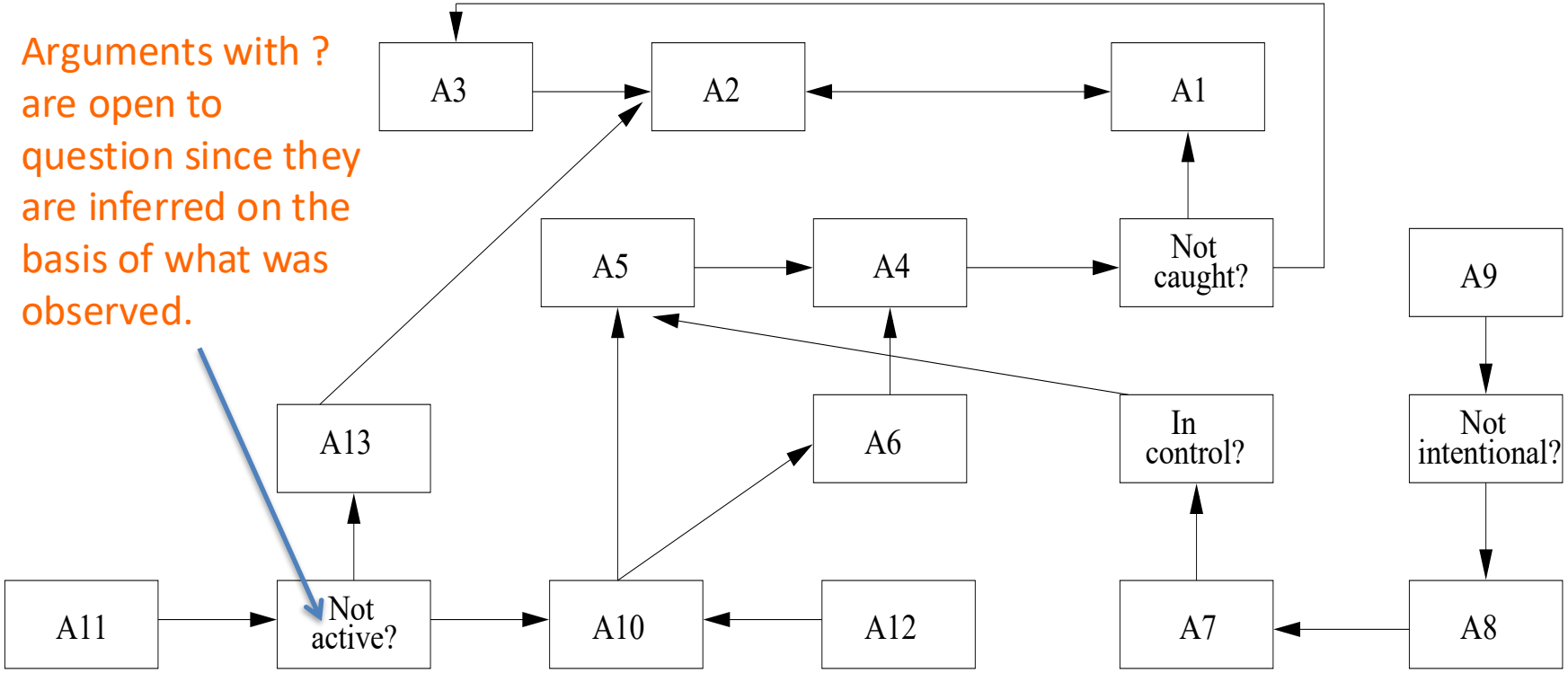
- The case as a computational model of argument



- Can make use of a number of different types of argument. Examples include:
  - *Argument from Witness Testimony (WT)*: Seventeen witnesses testified in the case, including Popov and Hayashi.
  - *Argument from Video Tape (VT)*: The video tape was used as evidence, and McCarthy relied on it at several points.
  - *Argument From Purpose (AP)*: McCarthy uses *teleological reasoning* in reaching his decision.

Argumentation Framework

Arguments with ? are open to question since they are inferred on the basis of what was observed.



Evaluation

- All arguments and attacks between them are evaluated.
- Judge McCarthy concludes that Popov has not caught the ball, but his attempt to do so is enough to establish a "*legally cognizable pre-possessory interest*" in the ball.
- This interest precludes Hayashi from establishing control by retrieving the ball.
- But, one of them must own it…



Evaluation

- So who owns the ball?



- McCarthy tries to come to a decision that is fair to both parties, invoking arguments from *purpose*.

- The result: McCarthy can decide neither for Popov nor for Hayashi.



- His final decision: the ball should be sold and the proceeds divided between the two.

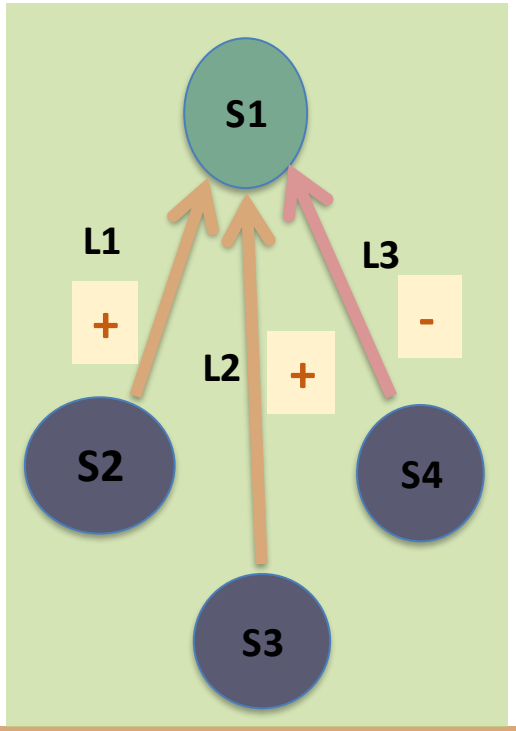


Handling Multiple Cases

- ANGELIC: ADF for kNowledGe Encapsulation of Legal Information from Cases
- A methodology for capturing knowledge of a legal domain, which is then used for deciding cases
  - Knowledge captured as an *Abstract Dialectical Framework* (Brewka et al.)
  - Operationalised in various implementations (Prolog, Java, Python)
- Evaluated on:
  - Three well known domains from the AI and Law literature
  - Multiple 'real world' domains supplied by law firm Weightmans
  - Cases from the European Court of Human Rights

Abstract Dialectical Frameworks

Links	Variety of interpretation
Nodes	Statement
Acceptance Condition	Different for each node



ADFs are a generalization of abstract Argumentation Frameworks (Dung 1995)

Definition:

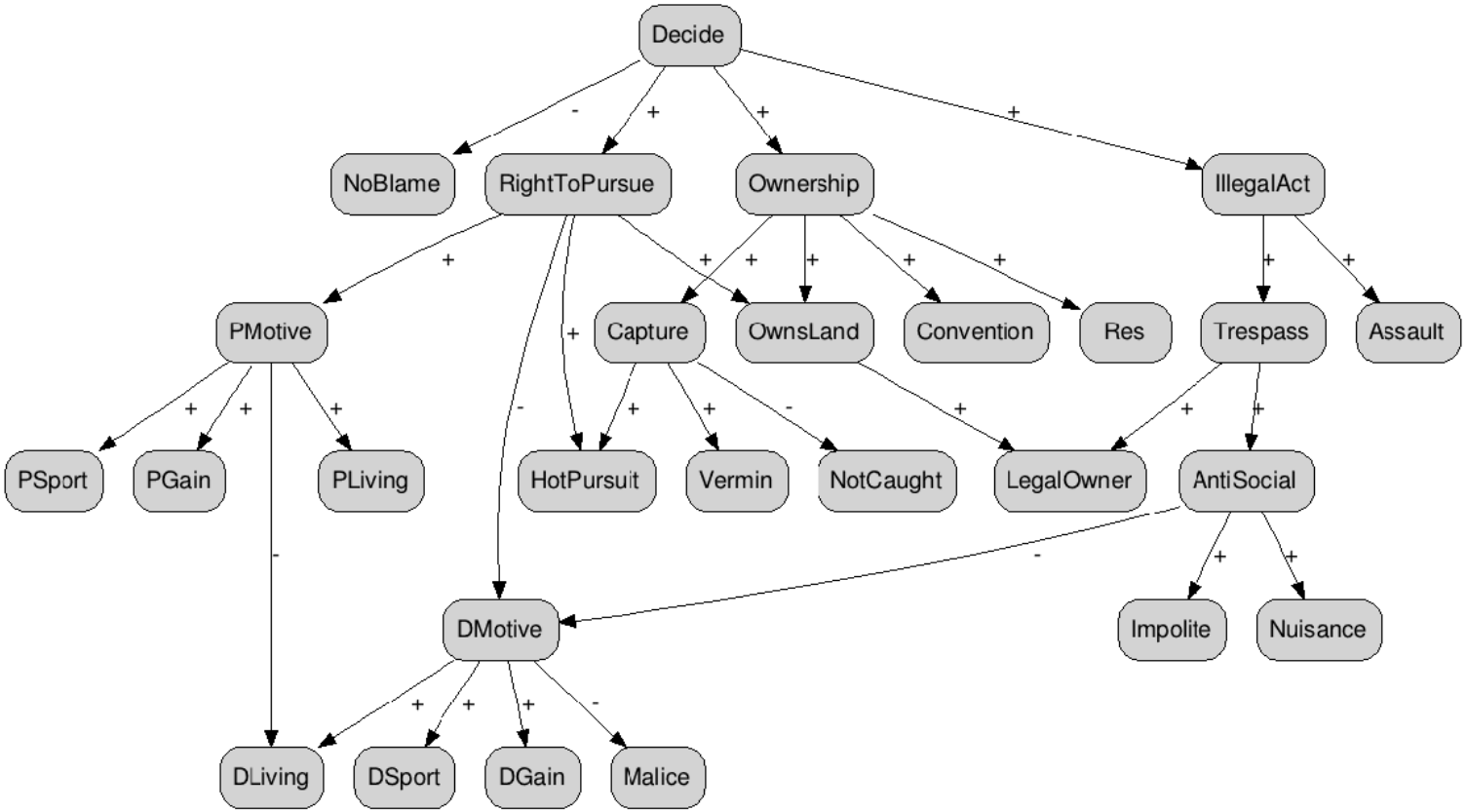
An ADF is a tuple **ADF=<S,L,C>** where

S is a set of statements (positions, nodes)

L is is a subset of *SxS* (where L can have + or – polarity)

C is an acceptance condition for *S*

Wild Animals domain model



Implementation

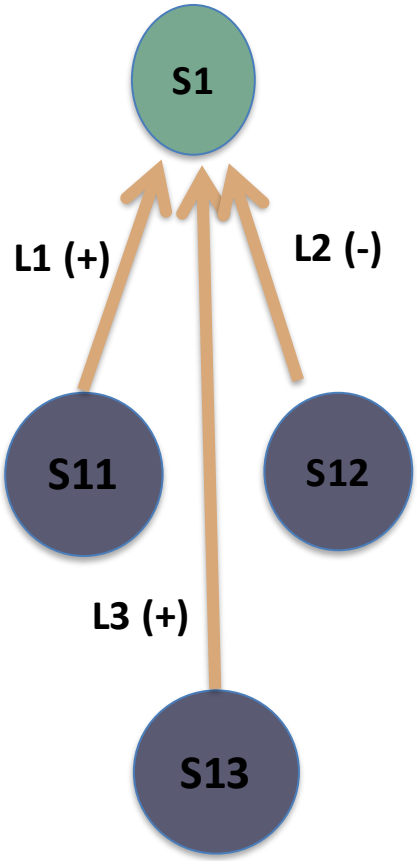
- We first constructed a complete ADF for the US trade secrets domain dataset
  - Prolog program produced with acceptance conditions captured as groups of clauses to determine the acceptability of each node in terms of its children
  - Captures 32 cases publicly available from the domain
- We repeated the exercise on two further domains
  - 5 cases in a domain on ‘wild animals’ (inc. baseball case)
  - 10 cases concerning the US automobile exception to the Fourth Amendment
    - Both well studied in AI and Law literature





Acceptance Conditions

Prolog Procedure



Prolog

Declarative semantics:

Facts:

Each case is represented by a set of base level factors **e.g. Case *C***

case(*C*, [S11, S12, S13]).

Rules:

```
(S1, present):- S11.              % L(+)
(S1, present):- S11 and S13.      % L(+)
(S1, absent):- S12.               % L(-)
(S1, default).    % Default: present or absent
```

Procedural semantics:

Ordering of clauses -> Priority

ECHR Domain

- Focus on Article 6 of the European Convention on Human Rights
  - Guarantees the right to a fair trial

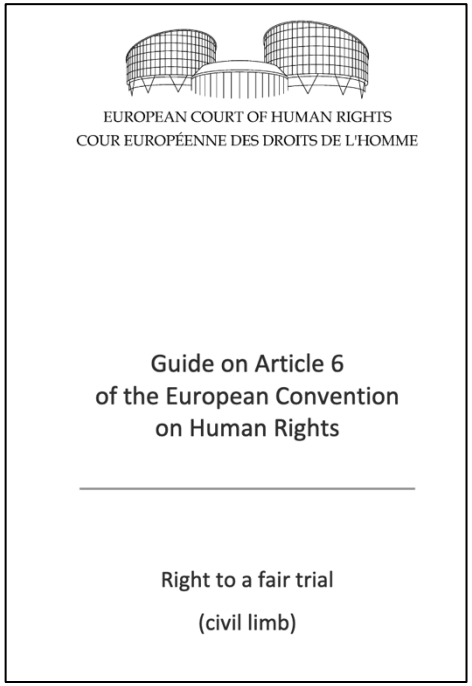
Article 6 § 1 of the Convention – Right to a fair trial

"1. In the determination of his civil rights and obligations or of any criminal charge against him, everyone is entitled to a fair and public hearing within a reasonable time by an independent and impartial tribunal established by law. Judgment shall be pronounced publicly but the press and public may be excluded from all or part of the trial in the interests of morals, public order or national security in a democratic society, where the interests of juveniles or the protection of the private life of the parties so require, or to the extent strictly necessary in the opinion of the court in special circumstances where publicity would prejudice the interests of justice. ..."

https://www.echr.coe.int/documents/guide_art_6_eng.pdf

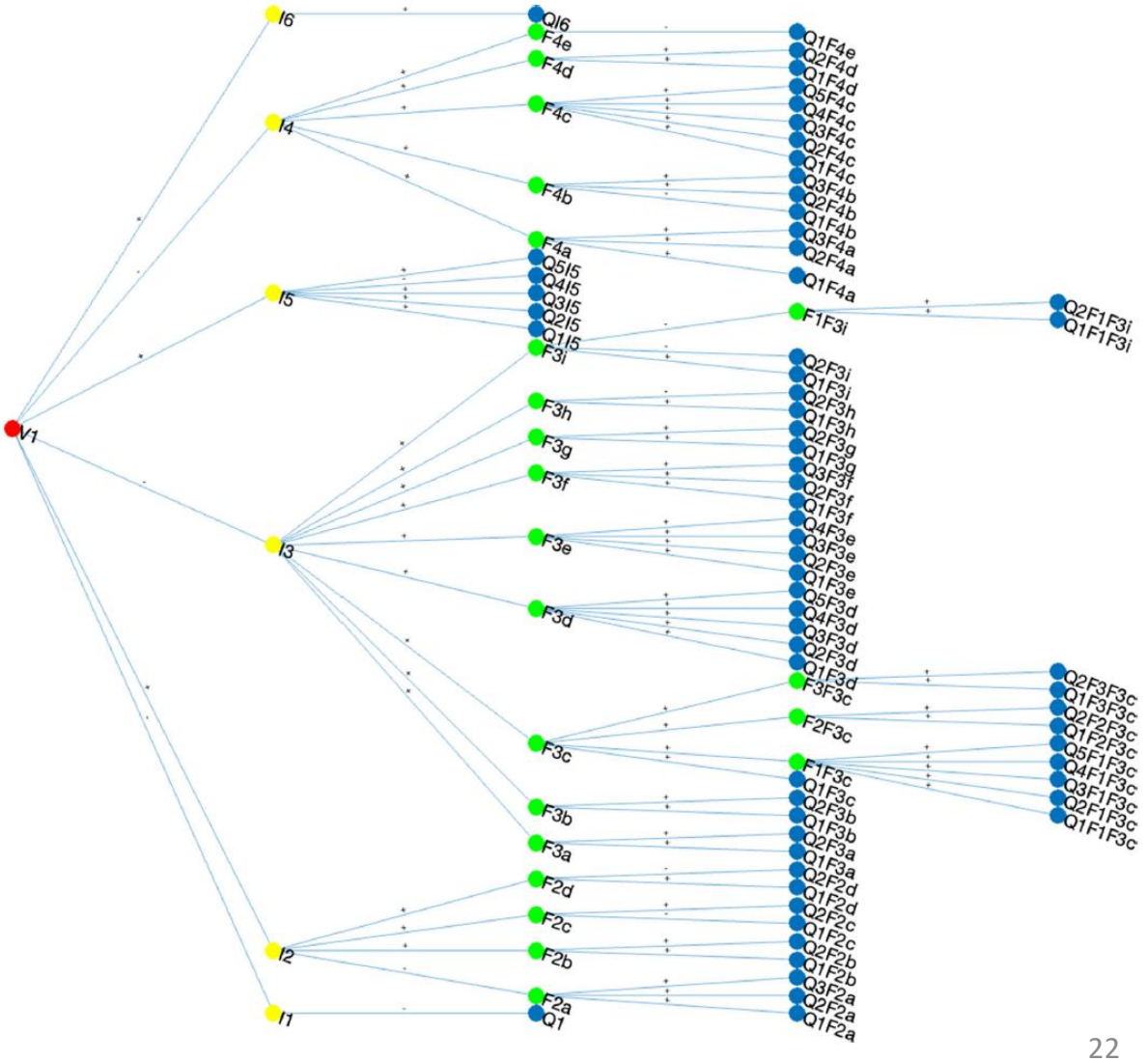
>70,000 pending applications!

115 pages



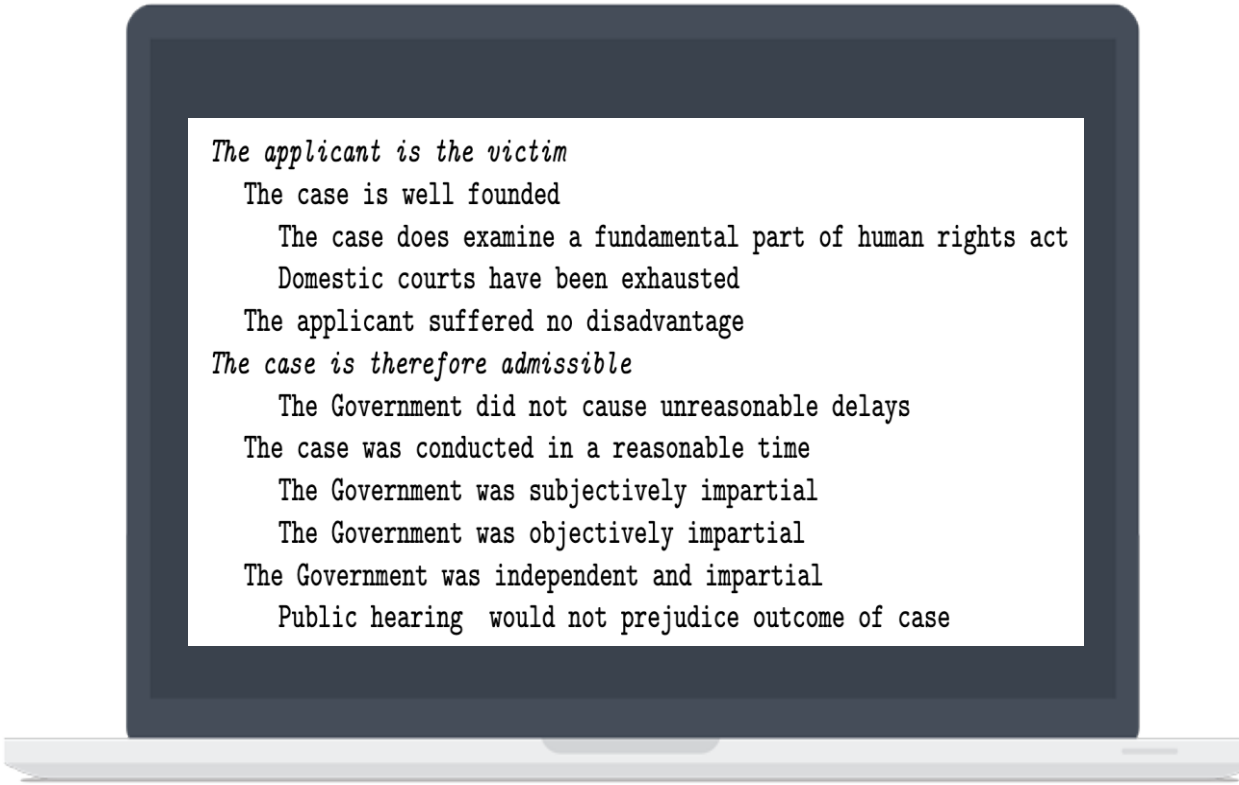
ECHR domain model

- The bottom of the ADF has base level factors: answers to questions given by the user on the basis of the facts of the case.
- Turned into an executable program by following the ANGELIC methodology



Article 6 Program Evaluation

- Accuracy
  - Tested on 10 cases from the ECHR dataset
    - 100% accuracy
  - Extended with another 30 cases
    - 97% accuracy



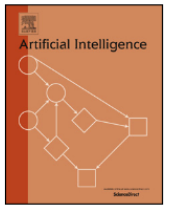
- Explainability
  - Program output for a specific case (Moser v. Austria)
  - Justifications for the outcomes evaluated by lawyers working in the ECtHR

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Explainable AI tools for legal reasoning about cases: A study on the European Court of Human Rights

Joe Collenette [a], Katie Atkinson [b,*], Trevor Bench-Capon [b]

[a] Department of Computer Science, University of Manchester, UK
[b] Department of Computer Science, University of Liverpool, UK

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ABSTRACT

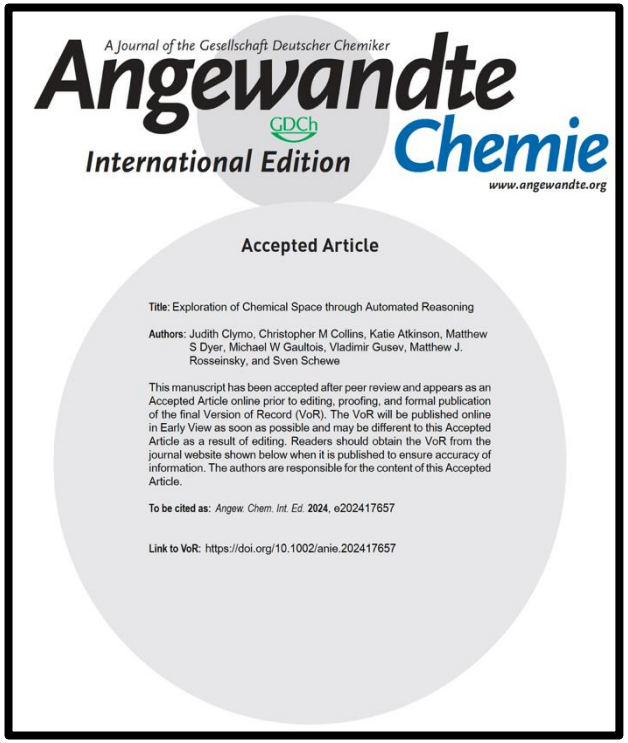
In this paper we report on a significant research project undertaken to design, implement and evaluate explainable decision-support tools for deciding legal cases. We provide a model of a legal domain, Article 6 of the European Convention on Human Rights, constructed using a methodology from the field of computational models of argument. We describe how the formal model has been developed, extended and transformed into practical tools, which were then used in evaluation exercises to determine the effectiveness and usability of the tools. The underpinning AI techniques used yield a level of explanation that is firmly grounded in legal reasoning and is also digestible by the target end users, as demonstrated through our evaluation activities. The results of our experimental evaluation show that on the first pass, our tool achieved an accuracy rate of 97% in matching the actual decisions of the cases and the user studies conducted gave highly encouraging results with respect to usability. As such, our project demonstrates how trustworthy AI tools can be built for a real world legal domain where critical needs of the end users are accounted for.

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And now for something completely different…

Explainable AI for Materials Discovery…..

Introducing Comgen



A Journal of the Gesellschaft Deutscher Chemiker

Angewandte GDCh **Chemie**

International Edition

www.angewandte.org

Accepted Article

Title: Exploration of Chemical Space through Automated Reasoning

Authors: Judith Clymo, Christopher M Collins, Katie Atkinson, Matthew S Dyer, Michael W Gaultois, Vladimir Gusev, Matthew J. Rosseinsky, and Sven Schewe

This manuscript has been accepted after peer review and appears as an Accepted Article online prior to editing, proofing, and formal publication of the final Version of Record (VoR). The VoR will be published online in Early View as soon as possible and may be different to this Accepted Article as a result of editing. Readers should obtain the VoR from the journal website shown below when it is published to ensure accuracy of information. The authors are responsible for the content of this Accepted Article.

To be cited as: *Angew. Chem. Int. Ed.* **2024**, e202417657

Link to VoR: https://doi.org/10.1002/anie.202417657

"*We report a tool, Comgen, for the efficient exploration of composition space, which makes use of logical methods from computer science used for proving theorems.*"

"Comgen accepts a variety of user-specified criteria, converts these into an abstract form, and utilises a powerful automated reasoning algorithm to identify compositions that satisfy these user requirements, or prove that the requirements cannot be simultaneously satisfied."

"In contrast to machine learning techniques, explicitly reasoning about domain knowledge, rather than making inferences from data, ensures that Comgen's outputs are fully interpretable and provably correct."

Boolean satisfiability



- Boolean formulas consist of variables that can take values 0 and 1 (for *false* and *true*); operators ∧, ∨, and ¬ (for *and*, *or*, and *negation*, respectively); and parentheses.

- Two example formulas:

  $\varphi = (x \vee y) \wedge (\neg x \vee \neg y)$

  $\psi = (x \vee y) \wedge \neg x \wedge \neg y$

- The Boolean satisfiability (SAT) problem asks, for a given Boolean formula, whether or not there is a satisfying assignment.

- φ has two satisfying assignments:

  x = 1, y = 0 and x = 0, y = 1;

  φ is therefore *satisfiable*.

- ψ has no satisfying assignment and is therefore unsatisfiable.



SAT for chemistry problems

- So how can we apply Boolean logic to chemistry problems?

- To find possible compositions, we may wish to restrict the choice of elements…

  Let the Boolean variable x_e mean:

  "the composition contains a non-zero quantity of element e".

Then we can express the requirement for a composition to contain an element from group one and an element from group seven as

$$(x_{Li} \vee x_{Na} \vee x_K \vee x_{Rb}) \wedge (x_F \vee x_{Cl} \vee x_{Br} \vee x_I)$$

- This reads as "*the composition contains a non-zero quantity of lithium or the composition contains a non-zero quantity of sodium or . . . and the composition contains a non-zero quantity of fluorine or . . .* "



SAT solvers

- To solve the satisfiability problem of a Boolean formula, we don’t want to take a naïve approach that tries all different combinations of valuations of the variables as this quickly becomes infeasible.
- Efficient ‘off-the-shelf’ SAT solvers are mature and available to direct at this task.

Constraints in Comgen

- Constraints captured in Comgen:
  - Quantity of an element
  - Charge balancing
  - Distance from known materials
  - Integer stoiciometry
  - Starting materials
  - Comparing properties of ions

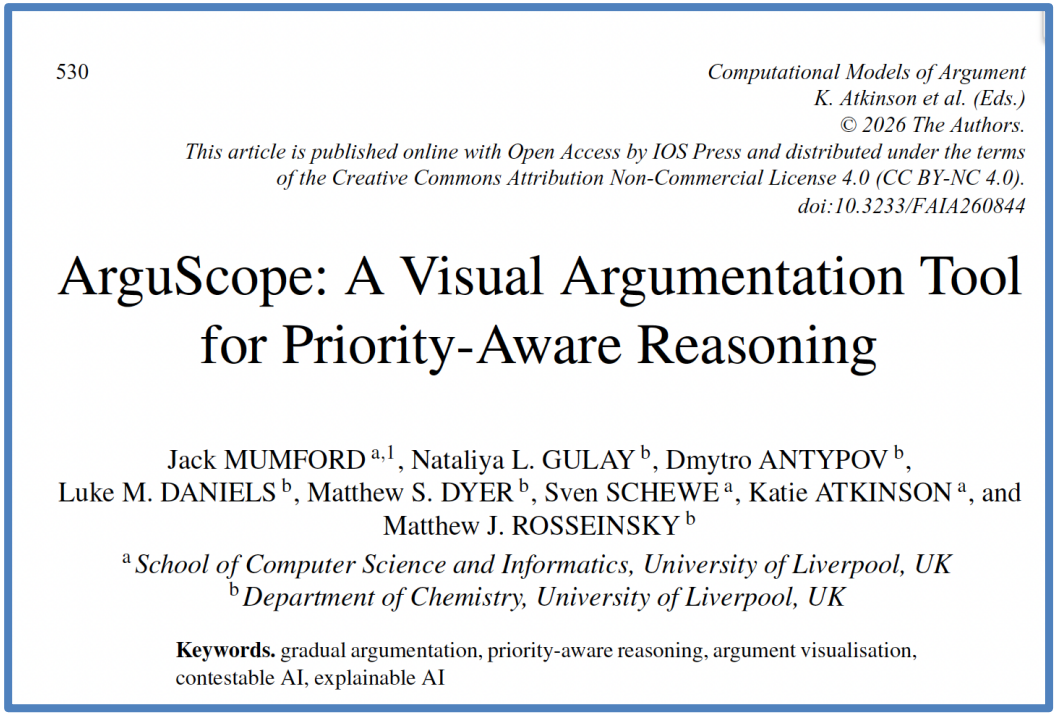


- Comgen is fronted by a high-level Python interface through which the user specifies which constraints are required and provide relevant parameters. The logical constraints are automatically generated and combined before being handed to the constraint solver used.

- Comgen is available as a Python package:
  https://pypi.org/project/comgen/

Adding in Arguments

- *ArguScope* has been developed to address the gap between formal argumentation models and the need to inspect, contest, and revise structured reasoning.



Computational Models of Argument
K. Atkinson et al. (Eds.)
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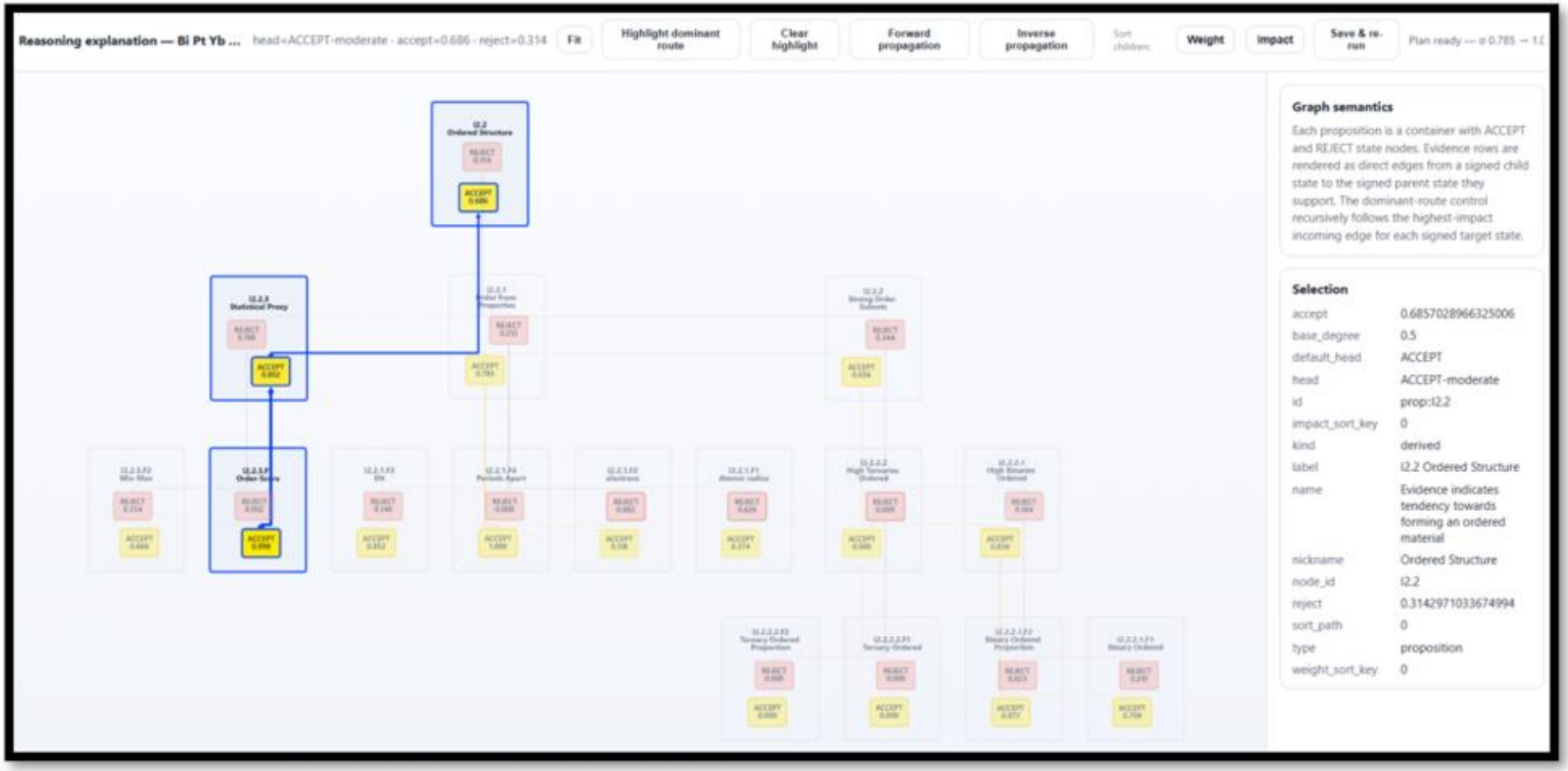
ArguScope: A Visual Argumentation Tool for Priority-Aware Reasoning

Jack MUMFORD [a,1], Nataliya L. GULAY [b], Dmytro ANTYPOV [b], Luke M. DANIELS [b], Matthew S. DYER [b], Sven SCHEWE [a], Katie ATKINSON [a], and Matthew J. ROSSEINSKY [b]

[a] *School of Computer Science and Informatics, University of Liverpool, UK*
[b] *Department of Chemistry, University of Liverpool, UK*

Keywords. gradual argumentation, priority-aware reasoning, argument visualisation, contestable AI, explainable AI

ArguScope Tool



Watch this space for functionality and application development of ArguScope…



Summary and Final Thoughts

- Computational models of argument bring high-fidelity explainability when used to model reasoning in domains where arguments can be expressed
- The knowledge acquisition bottleneck can be overcome by bringing other AI techniques to the development and instantiation of the argumentation models
- Domain reasoning is the important aspect to capture for explainability

Thank you for your attention!

Questions and comments?

References and Acknowledgements

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- Related publications can be found on my webpage: https://www.liverpool.ac.uk/people/katie-marie-atkinson/research-outputs